

Sequences & Patterns

ANSWER KEY



Arithmetic sequences

- 1 Find the 10th term of the sequence 5, 8, 11, 14, ...
 $a = 5, d = 3. a_{10} = 5 + 9 \times 3 = 32.$
 - 2 Find the 15th term of the sequence 7, 12, 17, 22, ...
 $a = 7, d = 5. a_{15} = 7 + 14 \times 5 = 77.$
 - 3 An arithmetic sequence has first term 4 and common difference 6. Find the 20th term.
 $a_{20} = 4 + 19 \times 6 = 4 + 114 = 118.$
 - 4 The 5th term of an arithmetic sequence is 23, and the first term is 3. Find the common difference.
 $3 + 4d = 23 \Rightarrow 4d = 20 \Rightarrow d = 5.$
 - 5 An arithmetic sequence has first term 7 and common difference 5. Which term equals 82?
 $7 + (n - 1) \times 5 = 82 \Rightarrow n - 1 = 15 \Rightarrow n = 16.$ The 16th term.
 - 6 The 8th term of an arithmetic sequence is 34, and the common difference is 4. Find the first term.
 $a + 7 \times 4 = 34 \Rightarrow a + 28 = 34 \Rightarrow a = 6.$
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Geometric sequences

- 7 Find the next term of the sequence 2, 4, 8, 16, ...
Geometric, ratio 2: $16 \times 2 = 32.$
 - 8 Find the 6th term of the sequence 3, 6, 12, 24, ...
 $a = 3, r = 2. a_6 = 3 \times 2^5 = 3 \times 32 = 96.$
 - 9 A geometric sequence has first term 5 and common ratio 3. Find the 5th term.
 $a_5 = 5 \times 3^4 = 5 \times 81 = 405.$
 - 10 The 4th term of a geometric sequence is 54, and the first term is 2. Find the common ratio.
 $2 \times r^3 = 54 \Rightarrow r^3 = 27 \Rightarrow r = 3.$
 - 11 A geometric sequence has 2nd term 6 and 3rd term 18. Find the first term and the common ratio.
 $r = 18 \div 6 = 3.$ First term $= 6 \div 3 = 2.$
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Identifying the type of sequence

- 12** Is 5, 10, 20, 40, ... arithmetic, geometric, or neither?
Each term is double the previous one ($r = 2$): geometric.
- 13** Is 6, 10, 14, 18, ... arithmetic, geometric, or neither?
Each term is 4 more than the previous one ($d = 4$): arithmetic.
- 14** Is 1, 4, 9, 16, ... arithmetic, geometric, or neither?
Differences 3, 5, 7 are not constant, and ratios are not constant either: neither (this is the sequence of perfect squares, n^2).
- 15** Is 100, 90, 81, 72.9, ... arithmetic, geometric, or neither?
Each ratio is $\frac{90}{100} = \frac{81}{90} = \frac{72.9}{81} = 0.9$: geometric with $r = 0.9$.
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Finding missing terms

- 16** Find the missing term: 3, __, 11, 15 (arithmetic).
Common difference $= (15 - 3) \div 3 = 4$. Missing term $= 3 + 4 = 7$.
- 17** Find the missing term: 4, __, 36 (geometric, all terms positive).
 $r = \sqrt{36 \div 4} = \sqrt{9} = 3$. Missing term $= 4 \times 3 = 12$.
- 18** Insert two arithmetic means between 5 and 20, forming the four-term sequence 5, __, __, 20.
Common difference $= (20 - 5) \div 3 = 5$. The sequence is 5, 10, 15, 20: the missing terms are 10 and 15.
- 19** The sequence x , $2x + 1$, $4x - 1$ is arithmetic. Find x .
Equal consecutive differences: $(2x + 1) - x = (4x - 1) - (2x + 1) \Rightarrow x + 1 = 2x - 2 \Rightarrow x = 3$.
- 20** Find the two missing terms of the geometric sequence 2, __, __, 54.
 $r^3 = 54 \div 2 = 27 \Rightarrow r = 3$. The sequence is 2, 6, 18, 54: the missing terms are 6 and 18.
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Sum of an arithmetic series

- 21** Find the sum of the first 10 terms of the sequence 2, 5, 8, 11, ...
 $a = 2$, $d = 3$, $n = 10$. $S = \frac{n}{2}(2a + (n - 1)d) = 5(4 + 27) = 5 \times 31 = 155$.
- 22** Find the sum of the first 20 positive integers.
 $S = \frac{20 \times 21}{2} = 210$.
- 23** Find the sum of the first 12 terms of the sequence 10, 7, 4, 1, ...
 $a = 10$, $d = -3$, $n = 12$. $S = 6(20 + 11 \times (-3)) = 6(20 - 33) = 6 \times (-13) = -78$.

- 24 An arithmetic sequence has first term 3 and common difference 4. Find the sum of its first 15 terms.
 $S = \frac{15}{2}(2 \times 3 + 14 \times 4) = 7.5 \times (6 + 56) = 7.5 \times 62 = 465.$
- 25 Find the sum of the first 25 positive even integers (2, 4, ..., 50).
 $a = 2, d = 2, n = 25. S = \frac{25}{2}(4 + 24 \times 2) = 12.5 \times 52 = 650.$

Special sequences

- 26 The sequence 1, 4, 9, 16, 25, ... consists of perfect squares. Find the 8th term.
 The n th term is n^2 , so the 8th term is $8^2 = 64.$
- 27 Find the 6th triangular number, from the sequence 1, 3, 6, 10, 15, ...
 The n th triangular number is $\frac{n(n+1)}{2}$. For $n = 6$: $\frac{6 \times 7}{2} = 21.$
- 28 A sequence starts 2, 3, 5, 8, 13, ..., where each term is the sum of the two terms before it. Find the next two terms.
 Next term = $8 + 13 = 21$. Following term = $13 + 21 = 34.$
- 29 Find the 7th term of the sequence of powers of 2: 1, 2, 4, 8, 16, ...
 The n th term is 2^{n-1} , so the 7th term is $2^6 = 64.$

Visual and growing patterns

- 30 The bar chart shows the number of small squares used to build patterns 1 through 4. Predict the number of squares in pattern 6, and state the rule.
 The rule is (pattern number)². Pattern 6 uses $6^2 = 36$ squares.
- 31 A pattern of dots grows as 3, 7, 11, 15, ... (each stage adds 4 dots). Find the number of dots in stage 10.
 $a = 3, d = 4. a_{10} = 3 + 9 \times 4 = 39.$
- 32 A row of matchstick triangles uses 3, 5, 7, 9, ... matchsticks for 1, 2, 3, 4 triangles. Find the number of matchsticks for 15 triangles in a row.
 $a = 3, d = 2. a_{15} = 3 + 14 \times 2 = 31.$
- 33 A row of matchstick hexagons uses 6, 11, 16, 21, ... matchsticks for 1, 2, 3, 4 hexagons. Find the matchsticks needed for 9 hexagons in a row.
 $a = 6, d = 5. a_9 = 6 + 8 \times 5 = 46.$

Word problems

- 34** A theater has 20 seats in the first row, and each following row has 3 more seats than the row before it. How many seats are in the 12th row?
 $a = 20, d = 3. a_{12} = 20 + 11 \times 3 = 53$ seats.
- 35** A ball is dropped and after each bounce rises to half its previous height. If it first rises to 80 m after being dropped, how high does it rise after the 4th bounce?
 Heights after each bounce form a geometric sequence with $a = 80, r = \frac{1}{2}$: after bounce 4, height
 $= 80 \times \left(\frac{1}{2}\right)^4 = 80 \div 16 = 5$ m.
- 36** A savings plan deposits 100 riyals in week 1, then increases the deposit by 20 riyals each following week. Find the total saved after 8 weeks.
 $a = 100, d = 20, n = 8. S = \frac{8}{2}(200 + 7 \times 20) = 4 \times 340 = 1,360$ riyals.
- 37** A bacteria culture starts with 50 bacteria and doubles every hour. How many bacteria are there after 6 hours?
 $50 \times 2^6 = 50 \times 64 = 3,200$ bacteria.

More on series and sequences

- 38** Find the sum of the first 8 terms of the geometric sequence 1, 2, 4, 8, ...
 $a = 1, r = 2. S = \frac{a(r^8 - 1)}{r - 1} = \frac{256 - 1}{1} = 255.$
- 39** A geometric sequence has first term 4 and common ratio 2. Find the sum of its first 6 terms.
 $S = \frac{4(2^6 - 1)}{2 - 1} = 4 \times 63 = 252.$
- 40** An arithmetic sequence has 3rd term 11 and 7th term 27. Find the common difference and the first term.
 $4d = 27 - 11 = 16 \Rightarrow d = 4.$ First term $= 11 - 2 \times 4 = 3.$
- 41** The n th term of a sequence is $T_n = 3n - 2$. Find the 12th term, and state whether the sequence is arithmetic.
 $T_{12} = 3(12) - 2 = 34.$ Since T_n is linear in n , the sequence is arithmetic, with common difference 3.

Pattern reasoning

- 42** A sequence of tile patterns is built so that stage n uses $n^2 + 1$ tiles. How many tiles are needed for stage 7?
 $7^2 + 1 = 49 + 1 = 50$ tiles.
- 43** The number of diagonals in a polygon with n sides is $\frac{n(n-3)}{2}$. Find the number of diagonals in an octagon ($n = 8$).
 $\frac{8 \times 5}{2} = 20$ diagonals.

- 44 A sequence follows the rule "double and add 1", starting at 3 as the 1st term. Find the 4th term.
Term 1 = 3. Term 2 = $2(3) + 1 = 7$. Term 3 = $2(7) + 1 = 15$. Term 4 = $2(15) + 1 = 31$.
- 45 The number of handshakes when n people each shake hands with every other person once is $\frac{n(n-1)}{2}$.
Find the number of handshakes among 10 people.
 $\frac{10 \times 9}{2} = 45$ handshakes.
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MCQ — Arithmetic sequences

- 46 Find the 12th term of the sequence 4, 9, 14, 19, ...

- a. 49
- b. 54
- c. 64
- d. 59 ✓

$$a_1 = 4, d = 5: T_{12} = 4 + 11(5) = 59.$$

- 47 Find the 18th term of the sequence 6, 13, 20, 27, ...

- a. 120
- b. 132
- c. 125 ✓
- d. 118

$$a_1 = 6, d = 7: T_{18} = 6 + 17(7) = 125.$$

- 48 An arithmetic sequence has first term 9 and common difference 4. Find the 25th term.

- a. 100
- b. 105 ✓
- c. 109
- d. 96

$$T_{25} = 9 + 24(4) = 105.$$

49 The 6th term of an arithmetic sequence is 41, and the first term is 6. Find the common difference.

a. 7 ✓

b. 8

c. 6

d. 35

$$41 = 6 + 5d \Rightarrow 5d = 35 \Rightarrow d = 7.$$

50 An arithmetic sequence has first term 10 and common difference 6. Which term equals 100?

a. 17th

b. 90th

c. 15th

d. 16th ✓

$$100 = 10 + (n - 1)(6) \Rightarrow n - 1 = 15 \Rightarrow n = 16.$$

51 The 10th term of an arithmetic sequence is 52, and the common difference is 5. Find the first term.

a. 12

b. 2

c. 7 ✓

d. 45

$$52 = a_1 + 9(5) \Rightarrow a_1 = 52 - 45 = 7.$$

MCQ — Geometric sequences

52 Find the next term of the sequence 3, 6, 12, 24, ...

- a. 30
- b. 27
- c. 36
- d. 48 ✓

Common ratio 2: $24 \times 2 = 48$.

53 Find the 7th term of the sequence 2, 6, 18, 54, ...

- a. 486
- b. 4,374
- c. 1,458 ✓
- d. 729

$a_1 = 2, r = 3: T_7 = 2(3^6) = 1,458$.

54 A geometric sequence has first term 4 and common ratio 2. Find the 6th term.

- a. 256
- b. 32
- c. 128 ✓
- d. 64

$T_6 = 4(2^5) = 128$.

55 The 5th term of a geometric sequence is 162, and the first term is 2. Find the common ratio.

- a. 3 ✓
- b. 9
- c. 4
- d. 2

$162 = 2r^4 \Rightarrow r^4 = 81 \Rightarrow r = 3$.

56 A geometric sequence has 3rd term 20 and 4th term 40. Find the first term and the common ratio.

a. $a_1 = 5, r = 2$ ✓

b. $a_1 = 20, r = 2$

c. $a_1 = 5, r = 4$

d. $a_1 = 10, r = 2$

$$r = 40 \div 20 = 2; a_1 = 20 \div r^2 = 20 \div 4 = 5.$$

57 A geometric sequence has first term 7 and common ratio 2. Find the 8th term.

a. 448

b. 1,792

c. 112

d. 896 ✓

$$T_8 = 7(2^7) = 896.$$

MCQ — Identifying the type of sequence

58 Is 7, 14, 28, 56, ... arithmetic, geometric, or neither?

a. Neither

b. Arithmetic

c. Geometric ✓

d. Both arithmetic and geometric

Each term is double the last (ratio 2): geometric.

59 Is 8, 13, 18, 23, ... arithmetic, geometric, or neither?

a. Both arithmetic and geometric

b. Geometric

c. Arithmetic ✓

d. Neither

Each term adds 5 (constant difference): arithmetic.

60 Is 2, 5, 10, 17, ... arithmetic, geometric, or neither?

a. Geometric

b. Arithmetic

c. Both arithmetic and geometric

d. Neither ✓

Differences are 3, 5, 7 (not constant) and ratios are not constant either: neither (it's $n^2 + 1$).

61 Is 200, 180, 162, 145.8, ... arithmetic, geometric, or neither?

a. Both arithmetic and geometric

b. Arithmetic

c. Geometric ✓

d. Neither

Each ratio is 0.9 (constant): geometric.

MCQ — Finding missing terms

62 Find the missing term: 6, __, 16, 21 (arithmetic).

a. 9

b. 13

c. 11 ✓

d. 10

$d = 5$ (from 16 to 21): missing = $16 - 5 = 11$.

63 Find the missing term: 5, __, 80 (geometric, all terms positive).

a. 42.5

b. 20 ✓

c. 16

d. 40

$r^2 = 80 \div 5 = 16 \Rightarrow r = 4$; missing = $5 \times 4 = 20$.

64 Insert two arithmetic means between 8 and 29, forming the four-term sequence 8, __, __, 29.

a. 15, 22 ✓

b. 14, 21

c. 12.5, 20.5

d. 15, 21

$d = \frac{29 - 8}{3} = 7$: terms are 15 and 22.

65 The sequence $2x$, $3x + 4$, $5x + 2$ is arithmetic. Find x .

a. 4

b. 2

c. 8

d. 6 ✓

$(3x + 4) - 2x = (5x + 2) - (3x + 4) \Rightarrow x + 4 = 2x - 2 \Rightarrow x = 6$.

66 Find the two missing terms of the geometric sequence 3, __, __, 81.

a. 12, 27

b. 9, 30

c. 27, 9

d. 9, 27 ✓

$r^3 = 81 \div 3 = 27 \Rightarrow r = 3$: terms are 9 and 27.

67 Find the missing term: __, 12, 19, 26 (arithmetic).

a. 7

b. 9

c. 4

d. 5 ✓

$d = 7$: missing = $12 - 7 = 5$.

MCQ — Sum of an arithmetic series

68 Find the sum of the first 12 terms of the sequence 3, 7, 11, 15, ...

a. 312

b. 300 ✓

c. 264

d. 282

$$S_{12} = \frac{12}{2}(2(3) + 11(4)) = 6(50) = 300.$$

69 Find the sum of the first 30 positive integers.

a. 900

b. 450

c. 435

d. 465 ✓

$$S = \frac{30(31)}{2} = 465.$$

70 Find the sum of the first 15 terms of the sequence 12, 9, 6, 3, ...

a. -135 ✓

b. -153

c. -108

d. 135

$$S_{15} = \frac{15}{2}(2(12) + 14(-3)) = 7.5(-18) = -135.$$

71 An arithmetic sequence has first term 5 and common difference 3. Find the sum of its first 20 terms.

a. 685

b. 670 ✓

c. 640

d. 625

$$S_{20} = \frac{20}{2}(2(5) + 19(3)) = 10(67) = 670.$$

72 Find the sum of the first 30 positive even integers (2, 4, ..., 60).

a. 465

b. 960

c. 900

d. 930 ✓

$$S = 30(31) = 930.$$

73 Find the sum of the first 10 terms of the sequence 20, 15, 10, 5, ...

a. -25 ✓

b. -50

c. 25

d. 5

$$S_{10} = \frac{10}{2}(2(20) + 9(-5)) = 5(-5) = -25.$$

MCQ — Sum of a geometric series

74 Find the sum of the first 6 terms of the sequence 2, 4, 8, 16, ...

a. 128

b. 124

c. 126 ✓

d. 62

$$S_6 = \frac{2(2^6 - 1)}{2 - 1} = 2(63) = 126.$$

75 A geometric sequence has first term 5 and common ratio 3. Find the sum of its first 5 terms.

a. 605.5

b. 1,210

c. 363

d. 605 ✓

$$S_5 = \frac{5(3^5 - 1)}{3 - 1} = \frac{5(242)}{2} = 605.$$

76 Find the sum of the first 4 terms of the sequence 1, 3, 9, 27, ...

a. 40 ✓

b. 81

c. 27

d. 39

$$1 + 3 + 9 + 27 = 40.$$

77 A geometric sequence has first term 6 and common ratio 2. Find the sum of its first 7 terms.

a. 768

b. 762 ✓

c. 384

d. 381

$$S_7 = \frac{6(2^7 - 1)}{2 - 1} = 6(127) = 762.$$

MCQ — Special sequences

78 The sequence 1, 4, 9, 16, 25, ... consists of perfect squares. Find the 10th term.

a. 100 ✓

b. 121

c. 90

d. 81

$$10^2 = 100.$$

79 Find the 7th triangular number, from the sequence 1, 3, 6, 10, 15, 21, ...

a. 28 ✓

b. 25

c. 36

d. 21

$$T_7 = \frac{7(8)}{2} = 28.$$

80 A sequence starts 3, 5, 8, 13, 21, ..., where each term is the sum of the two terms before it. Find the 8th term.

a. 55

b. 89 ✓

c. 144

d. 76

Continuing: 3, 5, 8, 13, 21, 34, 55, 89: the 8th term is 89.

81 Find the 8th term of the sequence of powers of 3: 1, 3, 9, 27, 81, ...

a. 729

b. 6,561

c. 2,184

d. 2,187 ✓

$$T_8 = 3^7 = 2,187.$$

MCQ — Visual and growing patterns

- 82 The bar chart shows the number of tiles used to build patterns 1 through 4. Predict the number of tiles in pattern 8.

- a. 37
- b. 40
- c. 29
- d. 33 ✓

$$d = 4, a_1 = 5: T_8 = 5 + 7(4) = 33.$$

- 83 A pattern of dots grows as 2, 6, 10, 14, ... (each stage adds 4 dots). Find the number of dots in stage 10.

- a. 38 ✓
- b. 40
- c. 34
- d. 42

$$a_1 = 2, d = 4: T_{10} = 2 + 9(4) = 38.$$

- 84 The bar chart shows the number of matchsticks used to build squares in a row, for 1 through 4 squares. Find the number needed for 12 squares.

- a. 34
- b. 43
- c. 40
- d. 37 ✓

$$d = 3, a_1 = 4: T_{12} = 4 + 11(3) = 37.$$

- 85 The bar chart shows the number of matchsticks used to build hexagons in a row, for 1 through 4 hexagons. Find the number needed for 9 hexagons.

- a. 41
- b. 51
- c. 50
- d. 46 ✓

$$d = 5, a_1 = 6: T_9 = 6 + 8(5) = 46.$$

MCQ — Word problems

- 86 A theater has 18 seats in the first row, and each following row has 4 more seats than the row before. How many seats are in row 15?

- a. 78
- b. 70
- c. 56
- d. 74 ✓

$$a_1 = 18, d = 4: T_{15} = 18 + 14(4) = 74.$$

- 87 A ball is dropped and after each bounce rises to half its previous height. If it first rises to 120 cm after the first bounce, how high does it rise after the 5th bounce?

- a. 60
- b. 15
- c. 3.75
- d. 7.5 ✓

$$a_1 = 120, r = 0.5: T_5 = 120(0.5)^4 = 7.5 \text{ cm.}$$

- 88 A savings plan deposits 80 riyals in week 1, then increases the deposit by 15 riyals each following week. Find the deposit in week 10.

a. 215 ✓

b. 200

c. 230

d. 935

$$a_1 = 80, d = 15: T_{10} = 80 + 9(15) = 215 \text{ riyals.}$$

- 89 A bacteria culture starts with 40 bacteria and triples every hour. How many bacteria are there after 4 hours?

a. 1,080

b. 3,240 ✓

c. 9,720

d. 480

$$40 \times 3^4 = 40 \times 81 = 3,240.$$

MCQ — Pattern reasoning

- 90 A sequence of tile patterns is built so that stage n uses $2n^2 + 1$ tiles. How many tiles are needed for stage 5?

a. 50

b. 26

c. 51 ✓

d. 41

$$2(25) + 1 = 51.$$

- 91 The number of diagonals in a polygon with n sides is $\frac{n(n-3)}{2}$. Find the number of diagonals in a polygon with 9 sides.

a. 36

b. 27 ✓

c. 21

d. 54

$$\frac{9(6)}{2} = 27.$$

- 92 A sequence follows the rule "triple and subtract 2", starting at 4 as the 1st term. Find the 4th term.

a. 34

b. 28

c. 82 ✓

d. 250

$4 \rightarrow 10 \rightarrow 28 \rightarrow 82$: the 4th term is 82.

- 93 The number of handshakes when n people each shake hands with every other person once is $\frac{n(n-1)}{2}$. Find the number of handshakes among 12 people.

a. 66 ✓

b. 132

c. 56

d. 78

$$\frac{12(11)}{2} = 66.$$

- 94 The n th term of a sequence is $T_n = 4n - 3$. Find the 15th term, and state whether the sequence is arithmetic or geometric.
- a. 57, geometric
 - b. 53, arithmetic
 - c. 60, arithmetic
 - d. 57, arithmetic ✓

$T_{15} = 4(15) - 3 = 57$; since T_n is linear in n , it's arithmetic.

- 95 The n th term of a sequence is $T_n = 5(2)^{n-1}$. Find the 6th term, and state whether the sequence is arithmetic or geometric.
- a. 160, geometric ✓
 - b. 320, geometric
 - c. 160, arithmetic
 - d. 80, geometric

$T_6 = 5(2^5) = 160$; since T_n involves a constant raised to the n th power, it's geometric.